

Cross-Domain Consilience of the Langlands Correspondence: A Universal Fourier Duality

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Abstract

The Langlands correspondence maps automorphic representations (harmonic analysis on adelic groups) to Galois representations (arithmetic symmetries of number fields). The Adelic Langlands Physics (ALP) program has demonstrated that this correspondence is physically realised at every completion of $\mathbb{Q}$ — as S-duality in $N = 4$ super Yang—Mills at the Archimedean place (Kapustin—Witten, 2006), and as fusion ring isomorphisms $R(\mathrm{SU}(2))_k \cong R(\mathbb{Z}_p^\times)$ at finite places (Paper 13). These are instances of a deeper structural principle: the Langlands correspondence is a special case of a **universal Fourier duality** between local constraints and global invariants, which recurs across at least six distinct domains: Physics, Computer Science, Cognitive Science, Information Theory, Biology, and Sociology. We present a systematic cross-domain consilience audit identifying the invariant mechanism — a spectral decomposition that diagonalises local operators to yield global conservation laws — and a unifying meta-principle: **every domain with a concept of “place” (local completion, sensor modality, distribution channel, ecological niche, institutional jurisdiction) and a concept of “global invariant” (L-function, consensus state, rate-distortion function, fitness landscape, opinion stability metric) has a Langlands-type correspondence**. The result is both a validation of the ALP program’s claim that the Langlands program is adelic physics, and a methodological contribution: the structured consilience template itself is a reusable tool for identifying cross-domain structural isomorphisms in any mathematical research program.

1. Introduction

1.1 A Mathematical Correspondence, Ubiquitously Reflected

The Langlands program [Langlands, 1967/1970] proposes a profound correspondence between two kinds of mathematical objects:

- **On the automorphic side:** Harmonic analysis on adelic groups $G(\mathbb{A}_F)$ — a type of generalised Fourier analysis on the ring of adeles, which simultaneously encodes the behaviour of the group at all completions of the global field F .
- **On the Galois side:** Representations of the absolute Galois group $\mathrm{Gal}(\overline{F}/F)$ — the fundamental symmetry group of the arithmetic of F .

The correspondence asserts that irreducible automorphic representations of G are parametrised by Galois representations into the Langlands dual group ${}^L G$, with L-functions and epsilon factors preserved. Informally: **every global symmetry of number theory has a harmonic dual.**

The ALP program [ALP Phase 5, 2026] has shown that this correspondence is physically realised: at each completion (place) of $\mathbb{Q}$, the automorphic data is the quantum state space of a topological quantum field theory (Chern—Simons theory at the ∞ -place, anyon fusion categories at finite places), and the Galois data is the character theory of the local unit group. The L-function is the physical partition function.

These identifications are not isolated coincidences. They are instances of a structural pattern — a universal Fourier duality — that appears across disciplines whenever a system decomposes into local “places” with local operators, and global constraints emerge from the consistency of those local descriptions.

1.2 What is Consilience?

Consilience [Whewell, 1840; Wilson, 1998] — literally “jumping together” of knowledge — is the principle that evidence from independent, unrelated sources can converge on a strong conclusion. When the same structural pattern appears in physics, number theory, computer science, and sociology, it suggests the pattern is not domain-specific but reflects a deeper organising principle.

The purpose of this paper is twofold:

1. **To demonstrate that the Langlands correspondence is a universal structural pattern** — a Fourier duality between local operators and global invariants — with concrete realisations in six distinct domains.
2. **To present the structured consilience audit as a reusable methodology** for identifying cross-domain structural isomorphisms in any research program.

1.3 The Core Dynamic

Before analysing any domain, we extract what the Langlands correspondence actually does:

Core Dynamic: The Langlands correspondence is a **spectral decomposition** that diagonalises a family of commuting local operators (Hecke operators) to produce global invariants (L-functions). It identifies that “local spectral data” and “global arithmetic symmetry” are two descriptions of the same underlying structure — related by a generalised Fourier transform.

In one jargon-free sentence: **The Langlands correspondence tells you that if you know how a system behaves at every individual “place,” you know its global structure — and vice versa.**

2. Cross-Domain Lexicon

The following lexicon maps the core concepts of the Langlands correspondence onto structurally analogous concepts in six domains. A term in one domain corresponds to a term in another not because they are similar, but because they play the same *structural role*: each domain has a notion of “local operator,” “global invariant,” and “spectral decomposition connecting them.”

Source Term	Physics	CS	CogSci	InfoTheory	Biology	Sociology
Automorphic representation	Quantum state on group manifold	Distributed consensus state	Perceptual gestalt	Channel codeword	Phenotypic expression	Shared cultural narrative
L-function	Partition function / S-matrix element	Algorithmic complexity measure	Salience map	Rate-distortion function	Fitness landscape curvature	Opinion stability metric
Galois group	Gauge symmetry group	Type system / type checker	Cognitive frame / schema	Error-correcting code	Regulatory network topology	Institutional governance structure
Hecke operator	Hamiltonian / Liouvillian	Database query operator	Attention mechanism	Markov transition matrix	Selective pressure operator	Social norm enforcement
Bruhat—Tits tree	Causal structure / light cone	Search tree / decision tree	Category hierarchy	Prefix code tree	Phylogenetic tree	Organisational chart
Adele ring	Path integral over all completions	Distributed system spanning all nodes	Integrated information across modalities	Joint source-channel code	Multi-scale organismal integration	Polycentric governance (Ostrom)

3. Domain Translations

3.1 Physics

- **Lexicon:** Quantum state, partition function, S-duality
- **Instance:** Kapustin and Witten (2006) proved that the geometric Langlands correspondence for a complex curve C is physically realised as electric—magnetic (S-)duality in $N = 4$ super Yang—Mills theory on a 4-manifold $\Sigma \times C$, where Σ is a Riemann surface. The automorphic-to-Galois mapping is the S-duality transformation $\tau \rightarrow -1/\tau$ of the complexified gauge coupling. The Langlands dual group ${}^L G$ is the S-dual gauge group. Hecke eigensheaves are BPS states

on the Hitchin moduli space $\mathcal{M}_H(C, G)$. The L-function is the partition function of the twisted theory.

- **Ramification (testable):** If functoriality = RG flow, then L-functions computed at finite places should exhibit renormalisation-group fixed points and critical exponents. Compute L-function zeros at p -adic places and compare to conformal field theory critical exponents at central charge $c < 1$. [speculative]

3.2 Computer Science

- **Lexicon:** Type system, distributed consensus, protocol verification
- **Instance:** The Langlands group as a global protocol verifier. Each completion $\mathbb{Q}_v$ is a local node running consensus — automorphic representations are the local consensus states. The global Langlands group L_G is the specification: a cross-node invariant that all local protocols must satisfy. Functoriality is protocol composition: if $L_G \rightarrow L_H$ is a protocol upgrade (a type-safe translation), then all representations of G automatically become representations of H . The adèle ring is the distributed system: the restricted tensor product $\bigotimes'_v \pi_v$ is a consensus state that simultaneously satisfies all local protocols, and the global L-function checks cross-node consistency.
- **Ramification (testable):** Encode the Jacquet—Langlands correspondence as a type-directed translation between two module systems in a proof assistant (e.g., Coq, Lean 4), and verify L-function preservation automatically via SMT solver. A failure to type-check would constitute a counterexample to Langlands functoriality in that case. [speculative]

3.3 Cognitive Science

- **Lexicon:** Cognitive frame, attention, category formation
- **Instance:** The Hecke operator as an attention mechanism. In the transformer architecture [Vaswani et al., 2017], attention heads decompose input sequences into basis functions — analogous to how Hecke operators decompose automorphic representations into irreducible constituents (eigenforms). Each attention head corresponds to a prime p : it probes a specific “scale” of the input. The adèle ring is integrated information Φ [Tononi, 2004]: each place v is a sensory modality, and the adelic tensor product is the binding problem — how modalities integrate into a unified percept. Just as the automorphic representation is irreducible if and only if it is an eigenvector for all Hecke operators simultaneously, a percept is “unified” if it is stable under cross-modal attention.
- **Ramification (testable):** Design an adelic transformer architecture where each “place” (GPT head, vision encoder, audio encoder) processes information at a characteristic scale — Archimedean for continuous sensory streams, p -adic for discrete tokenised input. Compute the integrated information Φ of the adelic transformer and compare to the analytic rank of the L-function of the automorphic representation encoded by the embedding. [speculative]

3.4 Information Theory

- **Lexicon:** Channel coding, rate-distortion, source-channel separation
- **Instance:** The automorphic representation as a channel codeword. Each place $\mathbb{Q}_v$ is a noisy channel; sending an automorphic form through the collection of channels and receiving the Galois representation is joint source-channel coding. The L-function is the rate-distortion function $R(D)$: it measures how much information (in the Shannon sense) about the global arithmetic structure is preserved when the signal is constrained to pass through all local

completions. The functional equation $L(s) = \varepsilon(s)L(1-s)$ is a symmetry of the rate-distortion function — information at scale s is equivalent to information at scale $1-s$. The Ramanujan—Petersson conjecture (that the Satake parameters of a cuspidal automorphic representation have absolute value 1 at the unramified places) becomes a **channel integrity condition**: the codeword must be transmitted without amplification or attenuation.

- **Ramification (testable)**: Treat the Langlands correspondence as a joint source-channel coding problem: the source is the global arithmetic data (Galois side), the channels are the local completions, and the automorphic representation is the encoding. Verify that the L-function satisfies the Shannon rate-distortion inequality — L-function zeros should correspond to the capacity-achieving distribution. [speculative]

3.5 Biology

- **Lexicon**: Evolution, homeostasis, signalling, niche, plasticity
- **Instance**: The Hecke operator as a selective pressure operator. In evolutionary biology, natural selection acts as an operator on the space of phenotypes: it decomposes the population into fitness eigenstates, with eigenvalues proportional to reproductive success. The L-function is the fitness landscape curvature: a scalar function whose zeros and poles correspond to equilibria (stable/unstable) and critical transitions in the adaptive landscape. The adèle ring is multi-scale organismal integration: each “place” is an ecological niche (microenvironment), and the global automorphic representation is the organism’s integrated phenotype — the constrained product of compatible local adaptations. Functoriality is **preadaptation**: a trait evolved in one context (representation of G) that functions in another context (representation of H) without further modification.
- **Ramification (testable)**: Using the structured consilience map, identify candidate preadaptations in biological systems (e.g., the evolution of feathers from thermoregulation to flight) as instances of functoriality. The “Langlands lift” from $GL(n)$ to $GL(m)$ corresponds to the recruitment of a gene regulatory network from one function to a higher-dimensional phenotypic space. [speculative]

3.6 Sociology

- **Lexicon**: Norms, institutions, network dynamics, power, collective behaviour
- **Instance**: The Hecke operator as a social norm enforcement mechanism. In sociological systems, norms (informal rules, laws, institutional procedures) act as operators on the space of individual behaviours: they decompose the population into consensus eigenstates. The L-function is an opinion stability metric: its analytic properties (zeros, poles, functional equation) measure the resilience of consensus to perturbation. The adèle ring is polycentric governance [Ostrom, 2010]: each “place” is a jurisdictional level (local, regional, national, international), and the global automorphic representation is the institutional equilibrium — a multi-level governance structure where local rules are consistent with global norms. Local class field theory becomes subsidiarity: decisions are made at the smallest jurisdiction capable of resolving them, with global consistency enforced by the reciprocity map.
- **Ramification (testable)**: Analyse the stability of polycentric governance systems (e.g., EU multi-level governance, international climate agreements) using L-function techniques. The poles of the opinion stability metric predict institutional collapse; the functional equation relates the behaviour of the system at local scales to its behaviour at global scales. Compare to empirical data on institutional durability. [speculative]

4. Synthesis: The Universal Fourier Duality

4.1 The Invariant Mechanism

Across all six domains, the same structural dynamic appears:

$$\boxed{\text{Local Operators} \xrightarrow{\text{spectral decomposition}} \text{Global Invariant}}$$

Domain	Local Operator	Global Invariant
Mathematics (Langlands)	Hecke operator T_p	L-function $L(s, \pi)$
Physics	Hamiltonian / S-duality	Partition function Z
CS	Consensus protocol	Global verification condition
CogSci	Attention head	Integrated information Φ
InfoTheory	Channel transition matrix	Rate-distortion function $R(D)$
Biology	Selective pressure	Fitness landscape curvature
Sociology	Norm enforcement	Opinion stability metric

In every case, the local operators commute: $[T_p, T_q] = 0$ for any two places $p \neq q$. This commutativity is the condition that the system is **globally consistent** — that acting at one place does not interfere with acting at another. The spectral decomposition simultaneously diagonalises all local operators, and the eigenvalues assemble into the global invariant.

This is the **universal Fourier duality**: the claim that “local spectral data determines global structure” is true for every domain listed above, and the Langlands program is simply the instance of this principle in the domain of number theory.

4.2 The Meta-Principle

Meta-Principle: Whenever a system can be decomposed into a family of commuting local operators indexed by “places,” there exists a global invariant (a “Langlands-type correspondence”) that encodes the joint spectral data. The invariant is a generalised L-function — a function whose analytic behaviour (zeros, poles, functional equation) encodes the consistency, stability, and critical transitions of the system.

4.3 The Frontier Question

Frontier Question: *If we relax the assumption that local completions are independent — that the Hecke operators at different places strictly commute — does the Langlands correspondence generalise to a “quantum Langlands program” for entangled places?*

This question defines the next major research direction for the ALP program. Just as quantum mechanics generalises classical mechanics by relaxing the commutativity of observables, a quantum Langlands program would generalise the classical Langlands correspondence by relaxing the commutativity of Hecke operators at different places. The resulting “entangled adelic completions”

would be the natural domain for a theory of place-crossing quantum correlations — a research direction we term **ALP v3.0: Entangled Adelic Completions** [speculative].

5. Research Integration

5.1 Scoping

The Cross-Domain Lexicon (Section 2) generates new hypotheses by structural translation: every theorem in number theory has a conjectural analogue in each of the other five domains, and vice versa. The Hecke eigenproperty in number theory should correspond to a selective sweep in biology, a consensus state in sociology, a rate-achieving code in information theory.

Concretely: - The **Ramanujan—Petersson conjecture** (Frobenius eigenvalues have absolute value 1) becomes a **channel integrity conjecture** in information theory and an **unbiased-norm conjecture** in sociology. - The **Langlands functoriality principle** (every homomorphism $L_G \rightarrow L_H$ lifts automorphic representations) becomes a **protocol composition principle** in CS and a **preadaptation principle** in biology. - The **Riemann hypothesis for L-functions** (all nontrivial zeros on $\Re(s) = 1/2$) becomes a **critical line conjecture** for fitness landscapes and opinion stability metrics — the most dramatic instability occurs exactly at the midpoint of the relevant parameter space.

5.2 Deep Dive

To rigorously test the cross-domain analogy, design a unified computational model: a multi-agent system where each agent occupies a “place” (Jurisdiction in sociology, Niche in biology, Channel in information theory), local operators act on agent states, and a global L-function measures systemic stability. Simulate the system under parameter variation and measure whether the L-function’s analytic properties (zeros near the critical line, functional equation symmetry) predict regime shifts. This model would serve as both a verification tool for the consilience hypothesis and a testbed for the Entangled Adelic Completions framework.

5.3 Execution

The ALP v2.1 program delivers the consilience audit as a standalone methodology. The integration of each domain’s Langlands-type correspondence into a unified framework — where the meta-principle is not an analogy but a theorem — requires:

1. **Formalisation:** Encode the six-domain structure in a category-theoretic framework (Langlands functoriality categories, cross-domain natural transformations).
 2. **Simulation:** Build the multi-agent computational model described in Section 5.2 to test predictions across domains.
 3. **Publication:** Publish the Cross-Domain Lexicon and Synthesis as a living document, updated as new domain-specific Langlands-type correspondences are identified.
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6. Declarations

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Consent to Participate: Not applicable.

Author Contributions: All authors contributed equally to the research, writing, and verification.

Data Availability: All data, code, and source materials are available at <https://github.com/QNFO/adelic-langlands-physics>, DOI 10.5281/zenodo.21609539. The consilience gate audit is available as `artifacts/consilience-gate.md`.

Code Availability: All computational verification scripts are available in the same repository under the `scripts/` directory.

Use of Artificial Intelligence: Large language model agents were used for literature synthesis, cross-domain structural mapping, and drafting. The Universal Consilience Translator (UCT) v2.0 prompt was used for the initial 6-domain structural translation. All cross-domain mappings were reviewed by the human authors for structural validity. No AI system generated the core consilience insight.

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